

# Methodological Advances in Microdata Imputation: Quantile Regression Forests for Wealth Imputation

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## Abstract

Policy analysts are regularly faced with the difficulty of performing policy analysis that relies on wealth without a comprehensive dataset capturing representative demographic, income and asset data across records. To address this challenge, this paper evaluates the methodological advantages of Quantile Regression Forests (QRF) for wealth imputation from the Survey of Consumer Finances (SCF) to the Current Population Survey (CPS), two US microdata surveys. We demonstrate that QRF outperforms traditional imputation approaches by preserving conditional distributions rather than merely conditional means, a critical distinction that makes distributional analyses of policy reforms under microdata more accurate. Our empirical analysis, implemented through the `microimpute` package, provides evidence that QRF reduces bias in wealth imputations, achieving a 20.5% reduction in average quantile loss (a measure of error against the full conditional distribution instead of central estimates) compared to OLS, 14.8% compared to Hot Deck Matching and a 6% reduction compared to Quantile Regression. Using 5-fold cross-validation on 22,975 SCF households, QRF achieves an average quantile loss across all quantiles of 6.6m, demonstrating superior distributional accuracy, particularly in the 10th-80th percentile range. While we focus on wealth in our experimental approach, these results suggest QRF’s advantages extend to any skewed variable requiring distributional preservation, including consumption, medical expenses, and other heavy-tailed economic measures. These technical improvements have substantial implications for microdata enhancement and fiscal policy research, enabling analysis from wealth tax values and asset-dependent SNAP or Medicaid qualifications to policy impact across wealth deciles. We release our open-source `microimpute` package to facilitate microimputation across the field, providing automated method comparison, hyperparameter tuning, and survey weight integration capabilities that streamline the imputation workflow for complex survey data.

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# 1 Introduction

Statistical matching, also known as data fusion or full variable imputation, addresses the challenge of combining information from multiple data sources that share common variables but contain different samples of units (D’Orazio et al., 2006), which is commonly found across fields in empirical research. In its simplest form, the problem involves a donor dataset containing variables  $(X, Y)$  and a receiver dataset containing only  $X$ , where the goal is to impute values of  $Y$  for observations in the receiver file based on their shared characteristics  $X$  (D’Orazio et al., 2021). This framework generalizes traditional missing data imputation by recognizing that “missingness” can arise not only from nonresponse within a single survey but also from the structural absence of variables across distinct data sources.

The relevance of statistical matching extends across the social sciences, particularly in microsimulation modeling where policy analysis requires combining detailed demographic information from one survey with economic variables from another (Bourguignon and Spadaro, 2006). For instance, household surveys may capture income and consumption patterns but lack wealth data, while financial surveys provide wealth information for different samples. Constructing synthetic files that combine these variables enables richer policy analysis than either source alone permits.

The methodological challenge lies in accurately modeling the conditional distribution  $P(Y|X)$  from the donor data and using it to generate plausible values of  $Y$  for receiver observations. Different approaches make different assumptions about this distribution. Parametric methods like Ordinary Least Squares assume linearity and homoscedasticity, which are often violated by economic variables exhibiting heavy tails and heterogeneous relationships across the distribution (Meinshausen and Ridgeway, 2006). Nonparametric methods like hot deck matching avoid distributional assumptions but may struggle to capture complex multivariate relationships (Andridge and Little, 2010). More recent machine learning approaches, including Quantile Regression Forests and Mixture Density Networks, offer flexible alternatives that can model entire conditional distributions without restrictive parametric assumptions.

This paper presents the `microimpute` package, a Python library implementing five statistical matching methods: Hot Deck Matching, Ordinary Least Squares, Quantile Regression, Mixture Density Networks, and Quantile Regression Forests. The package provides a comprehensive framework for comparing these methods through systematic benchmarking, with automated model selection based on quantile loss performance. We demonstrate the methodology through an application to wealth imputation, transferring net worth data from the Survey of Consumer Finances to the Current Population Survey. This task exemplifies the challenges of statistical matching for heavy-tailed distributions.

This work provides the following contributions:

- An open-source implementation of a unified framework for five imputation methods, facilitating systematic comparison across methods and support statistical matching through accessible, user-friendly software
- An automated benchmarking pipeline that evaluates distributional accuracy using quantile loss, enabling researchers to select the most appropriate method for their specific data characteristics

- Empirical evidence on the relative performance of these methods for wealth imputation, confirming the advantages of Quantile Regression Forests for heavy-tailed distributions

The remainder of this paper is organized as follows. Section 2 reviews the statistical matching problem, including its formal definition, the conditional independence assumption that underlies all matching methods, and the five imputation approaches implemented in `microimpute`. Section 3 describes our data sources. Section 4 presents the package architecture and benchmarking methodology. Section 5 reports empirical results from the wealth imputation application. Section 6 discusses implications and limitations, and Section 7 concludes.

## 2 Background

This section establishes the theoretical foundations for statistical matching and reviews the imputation methods implemented in `microimpute`. We begin with the formal problem definition and the key assumption underlying all statistical matching procedures, then discuss the relationship to missing data mechanisms, and finally describe each of the five imputation methods.

### 2.1 The Statistical Matching Problem

Statistical matching arises when a researcher seeks to combine information from two (or more) data sources that have been collected independently on different samples (D’Orazio et al., 2006). The canonical setup involves:

- A **donor file**  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n_D}$  containing  $n_D$  observations with common variables  $X$  and target variable(s)  $Y$
- A **receiver file**  $\mathcal{R} = \{x_j\}_{j=1}^{n_R}$  containing  $n_R$  observations with only the common variables  $X$

The goal is to impute values  $\hat{y}_j$  for each observation  $j$  in the receiver file, creating a synthetic file that combines the variables from both sources. When the receiver file also contains variables  $Z$  not present in the donor file, the resulting synthetic file contains all three sets of variables  $(X, Y, Z)$ , though  $Y$  and  $Z$  are never jointly observed.

Formally, statistical matching seeks to estimate the conditional distribution  $P(Y|X)$  from the donor data and use it to generate imputations for the receiver observations. Let  $\hat{P}(Y|X = x)$  denote an estimate of this conditional distribution. For a receiver observation with covariates  $x_j$ , we can either:

1. Predict a specific quantile, such as the conditional median:  $\hat{y}_j = \hat{Q}_{0.5}(Y|X = x_j)$
2. Draw a random value from the estimated distribution:  $\hat{y}_j \sim \hat{P}(Y|X = x_j)$

The first approach (deterministic imputation) may be appropriate when only point predictions are needed, though it systematically underestimates variance (Rubin, 1987). The second approach (stochastic imputation) preserves the variability of  $Y$  and is preferred when distributional properties must be maintained. This is often the case in microsimulation modeling, where full distributional policy analysis is required.

## 2.2 The Conditional Independence Assumption

All statistical matching procedures rely, either explicitly or implicitly, on the Conditional Independence Assumption (CIA):

$$Y \perp Z \mid X \tag{1}$$

This assumption states that the target variable  $Y$  (from the donor) and any variables  $Z$  unique to the receiver are independent, conditional on the common variables  $X$  (D’Orazio et al., 2006). In other words, once we account for the shared covariates, knowing  $Z$  provides no additional information about  $Y$ .

The CIA is fundamentally untestable because  $Y$  and  $Z$  are never jointly observed in either data source. Its plausibility depends on the richness of the common variables  $X$ . When  $X$  captures the key determinants of both  $Y$  and  $Z$ , the assumption is more likely to hold. In practice, researchers must rely on substantive knowledge to assess whether the available common variables are sufficient to render  $Y$  and  $Z$  conditionally independent.

Violations of the CIA lead to biased estimates of the joint distribution of  $(Y, Z)$  in the synthetic file. However, if the research question concerns only the marginal distribution of  $Y$  in the receiver population, the CIA is not strictly required. What matters is that the conditional distribution  $P(Y|X)$  is correctly specified and that  $X$  has sufficient overlap between donor and receiver files (D’Orazio et al., 2021).

## 2.3 Missing Data Mechanisms

Statistical matching may also warrant discussion within the framework of missing data theory, where  $Y$  is “missing” for all observations in the receiver file (Little and Rubin, 2002). The missing data literature distinguishes three mechanisms:

- **Missing Completely at Random (MCAR):** The probability of missingness is unrelated to both observed and unobserved variables. In the statistical matching context, this would require that selection into the donor versus receiver sample is independent of all variables.
- **Missing at Random (MAR):** The probability of missingness depends only on observed variables. For statistical matching, this means that conditional on  $X$ , the probability of being in the donor sample (and thus having  $Y$  observed) does not depend on the value of  $Y$  itself.
- **Missing Not at Random (MNAR):** The probability of missingness depends on the unobserved values. This would occur if, for example, high-wealth individuals were systematically less likely to appear in the wealth survey.

Under MAR, valid inference about  $P(Y|X)$  can be obtained from the donor data alone, which justifies the statistical matching approach. Most imputation methods, including those implemented in `microimpute`, assume MAR. When MNAR is suspected, sensitivity analyses or methods that explicitly model the selection mechanism may be required (Little and Rubin, 2002).

In statistical matching applications, an important consideration arises when the donor and receiver files come from surveys with different sampling designs. For instance, wealth surveys often oversample high-net-worth households to improve precision for this rare population, while general household surveys use designs representative of the broader population. This differential sampling can introduce a form of selection that resembles MNAR if not properly addressed. Luckily, survey weights provide a mechanism to mitigate this issue. By incorporating weights that reflect each observation’s representativeness of the target population, imputation methods can adjust for known differences in sampling probabilities between the donor and receiver surveys (DuGoff et al., 2014). The `microimpute` package supports the use of survey weights throughout its imputation pipeline, allowing methods to account for complex survey designs. While survey weights cannot fully address MNAR when missingness depends on unobserved values of  $Y$  itself, they can substantially reduce bias arising from differential sampling designs across data sources.

## 2.4 Imputation Methods

We now describe the five imputation methods implemented in `microimpute`. Each method provides a different approach to estimating the conditional distribution  $P(Y|X)$  and generating imputations.

### 2.4.1 Hot Deck Matching

Hot deck imputation replaces missing values in a receiver record with observed values from “similar” donor records (Andridge and Little, 2010). The `microimpute` implementation uses the constrained distance hot deck approach from the StatMatch R package (D’Orazio et al., 2021). To identify the best match, the method computes distances between each receiver observation and all donor observations based on the common variables  $X$ , selects donor records within a specified distance threshold, and randomly samples from the eligible donors, with optional weighting by survey weights. For continuous and mixed covariates, Mahalanobis or Gower distance metrics can capture similarity across different variable types. The donated value will thus be an actual observed value from the donor file, ensuring plausibility.

Hot deck methods are nonparametric and avoid distributional assumptions, making them robust when the true conditional distribution is unknown or complex (D’Orazio et al., 2006). However, they face limitations:

- **Donor scarcity:** For receiver observations in sparse regions of the covariate space suitable donors may be unavailable, which can lead to the reuse of the same donors, which can affect the accuracy in the distribution of imputed values
- **Discrete outputs:** Imputed values are restricted to the set of observed donor values, which may not adequately represent the continuous nature of  $Y$

- **Tail behavior:** Extreme values in the receiver population may have no appropriate donors or poor matches, leading to biased imputations at the tails

### 2.4.2 Ordinary Least Squares (OLS)

OLS imputation models the conditional mean of  $Y$  given  $X$  through linear regression:

$$E[Y|X] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p \quad (2)$$

The coefficients  $\beta$  are estimated from the donor data by minimizing squared residuals. For stochastic imputation, the `microimpute` implementation, which builds on the `statsmodels` library, assumes normally distributed errors:

$$Y|X \sim \mathcal{N}(\mathbf{X}'\hat{\beta}, \hat{\sigma}^2) \quad (3)$$

where  $\hat{\sigma}^2$  is the estimated residual variance. Quantiles of  $Y|X$  can then be computed as:

$$\hat{Q}_\tau(Y|X) = \mathbf{X}'\hat{\beta} + \hat{\sigma} \cdot \Phi^{-1}(\tau) \quad (4)$$

where  $\Phi^{-1}(\tau)$  is the  $\tau$ -th quantile of the standard normal distribution.

OLS is computationally efficient and well-understood, but its assumptions are often violated by economic variables:

- **Linearity:** Relationships between  $Y$  and  $X$  may be nonlinear
- **Homoscedasticity:** Variance often increases with the level of  $Y$ , particularly for wealth
- **Normality:** Heavy-tailed distributions violate the normal error assumption, leading to inaccurate quantile estimates (Von Hippel, 2007)

### 2.4.3 Quantile Regression

Quantile regression directly models conditional quantiles rather than the conditional mean (Koenker and Bassett, 1978). For quantile  $\tau \in (0, 1)$ , the model is:

$$Q_\tau(Y|X) = \beta_0(\tau) + \beta_1(\tau)x_1 + \cdots + \beta_p(\tau)x_p \quad (5)$$

The coefficients  $\beta(\tau)$  are estimated by minimizing the asymmetric quantile loss function:

$$\mathcal{L}_\tau(\beta) = \sum_{i=1}^n \rho_\tau(y_i - \mathbf{x}'_i \beta) \quad (6)$$

where the check function  $\rho_\tau(u) = u(\tau - \mathbf{1}_{u < 0})$  penalizes positive and negative residuals asymmetrically.

The `microimpute` implementation of Quantile regression also uses the `statsmodels` library. By fitting models at multiple quantiles, it can approximate the entire conditional distribution of  $Y|X$ . For imputation, a random quantile  $\tau^* \sim \text{Uniform}(0, 1)$  can be drawn, and the imputed value computed by interpolating between estimated quantile functions (Wei et al., 2014).

Quantile regression offers advantages over OLS:

- **Robustness:** Less sensitive to outliers than least squares
- **Heteroscedasticity:** Naturally accommodates varying spread across the distribution
- **No normality assumption:** Does not require specifying an error distribution

However, quantile regression still assumes linear relationships at each quantile, limiting its flexibility for complex data structures (Meinshausen and Ridgeway, 2006).

#### 2.4.4 Mixture Density Networks (MDN)

Mixture Density Networks combine neural networks with mixture models to estimate flexible conditional densities (Bishop, 1994). The approach models  $P(Y|X)$  as a mixture of Gaussian components:

$$p(y|x) = \sum_{k=1}^K \pi_k(x) \cdot \mathcal{N}(y | \mu_k(x), \sigma_k^2(x)) \quad (7)$$

where  $K$  is the number of mixture components, and the mixing coefficients  $\pi_k(x)$ , means  $\mu_k(x)$ , and variances  $\sigma_k^2(x)$  are all functions of the input  $x$ , parameterized by a neural network.

The neural network outputs  $3K$  values for each input:  $K$  unnormalized log-probabilities (transformed via softmax to obtain  $\pi_k$ ),  $K$  means, and  $K$  log-variances (exponentiated to ensure positivity). The network is trained by maximizing the log-likelihood:

$$\mathcal{L} = \sum_{i=1}^n \log p(y_i|x_i) \quad (8)$$

For imputation, one can either sample from the mixture distribution or compute quantiles numerically by inverting the cumulative distribution function.

The `microimpute` implementation uses PyTorch Tabular (Joseph, 2021), which provides optimized architectures for tabular data. MDNs offer several advantages:

- **Multimodal distributions:** The mixture formulation can capture multiple modes in  $P(Y|X)$
- **Flexible nonlinearity:** Neural networks can approximate complex relationships between  $X$  and distributional parameters
- **Full density estimation:** Provides a complete probabilistic model, not just quantiles

Nonetheless, its limitations include:

- **Data requirements:** Neural networks typically require larger sample sizes for reliable estimation
- **Hyperparameter sensitivity:** Performance depends on architecture choices, learning rate, and number of components
- **Computational cost:** Training is more expensive than parametric methods

### 2.4.5 Quantile Random Forests (QRF)

Quantile Random Forests extend Random Forests to estimate conditional quantiles (Meinshausen and Ridgeway, 2006). A standard Random Forest builds an ensemble of decision trees, each trained on a bootstrap sample, with predictions averaged across trees. QRF modifies this by retaining all training observations in each terminal node rather than just their mean.

For a new observation with covariates  $x$ , QRF estimates the conditional distribution function as:

$$\hat{F}(y|X = x) = \sum_{i=1}^n w_i(x) \cdot \mathbf{1}_{Y_i \leq y} \quad (9)$$

where the weight  $w_i(x)$  reflects how often training observation  $i$  falls in the same terminal node as  $x$  across trees in the forest. Specifically:

$$w_i(x) = \frac{1}{B} \sum_{b=1}^B \frac{\mathbf{1}_{x_i \in L_b(x)}}{|L_b(x)|} \quad (10)$$

where  $B$  is the number of trees,  $L_b(x)$  is the terminal node containing  $x$  in tree  $b$ , and  $|L_b(x)|$  is the number of training observations in that node.

The  $\tau$ -th conditional quantile is then:

$$\hat{Q}_\tau(Y|X = x) = \inf\{y : \hat{F}(y|X = x) \geq \tau\} \quad (11)$$

The `microimpute` implementation uses the `quantile-forest` package, which offers several advantages for statistical matching:

1. **Distribution preservation:** By modeling the entire conditional distribution, QRF captures skewness, heavy tails, and other distributional features without parametric assumptions (Meinshausen and Ridgeway, 2006).
2. **Nonlinear relationships:** The tree-based structure automatically captures complex nonlinearities and interactions without requiring explicit specification (Breiman, 2001).
3. **Single model for all quantiles:** Unlike standard quantile regression, which requires fitting separate models for each quantile, QRF estimates all quantiles from a single trained forest.
4. **Robustness:** The ensemble approach and quantile focus make QRF less sensitive to outliers than mean-based methods (Tang and Ishwaran, 2017).

Nevertheless, QRF also has limitations:

- **Extrapolation:** Trees cannot extrapolate beyond the range of training data, potentially underestimating extreme quantiles when donor and receiver distributions differ
- **Discrete approximation:** The estimated distribution is discrete, with support limited to observed  $Y$  values in the training data
- **Computational cost:** Training and prediction are slower than parametric methods, though generally faster than neural networks



## 2.5 Current Practice in Microsimulation

Statistical matching and data fusion are fundamental operations in microsimulation modeling, yet the methods employed in practice have remained relatively unchanged for decades. A review of major tax-benefit microsimulation models reveals a strong reliance on traditional imputation approaches, primarily hot deck matching and OLS-based regression.

Hot deck matching remains the dominant approach in European microsimulation. EU-ROMOD, the EU-wide tax-benefit model, employs a multi-stage imputation procedure combining predictive mean matching with distance-based hot deck methods to integrate consumption data from Household Budget Surveys into its EU-SILC input data (Sutherland and Figari, 2013). The appeal of hot deck methods lies in their simplicity and the guarantee that imputed values are observed values from the donor file, ensuring plausibility. However, as discussed above, these methods struggle with tail behavior and may not adequately capture the full conditional distribution of the target variable.

In U.S. tax policy microsimulation, regression-based approaches are more common. The Tax Policy Center employs a two-stage probit and OLS procedure for wealth imputation, first predicting the probability of holding each asset type, then predicting amounts conditional on positive holdings (Nunns, 2012). Similarly, the Institute on Taxation and Economic Policy (ITEP) model relies on statistical matching between tax return data and the American Community Survey, supplemented with regression-based imputations from the Survey of Consumer Finances (Institute on Taxation and Economic Policy, 2023). These regression approaches assume linear relationships and normally distributed errors; assumptions that are frequently violated by economic variables with heavy tails and heteroscedastic relationships.

More recent microsimulation efforts have begun incorporating administrative data through record linkage rather than statistical matching (Abowd, 2018), but this approach requires access to restricted data and raises privacy concerns. For researchers working with publicly available survey data, statistical matching remains essential, creating a need for methods that can better capture complex distributional features.

The limitations of traditional approaches can be particularly acute for heavily-tailed variables where relationships with predictors vary across the distribution. Machine learning methods such as Quantile Regression Forests offer a promising alternative, as they can capture nonlinear relationships, model the entire conditional distribution, and handle heavy-tailed data without restrictive parametric assumptions. Despite these advantages, QRF and similar methods have seen limited adoption in mainstream microsimulation practice. The `microimpute` package aims to lower barriers to adopting these more flexible methods by providing a unified framework for comparing traditional and machine learning approaches, enabling researchers to empirically evaluate which method best suits their specific data characteristics.

## 3 Data

We demonstrate the `microimpute` methodology through an application to wealth imputation, transferring net worth from the Survey of Consumer Finances (SCF) onto the Current Population Survey (CPS).

### 3.1 Survey of Consumer Finances

The Survey of Consumer Finances (SCF), sponsored by the Federal Reserve Board, is a triennial survey providing detailed information on U.S. households’ assets, liabilities, income, and demographic characteristics. Its dual-frame sample design includes a standard national area-probability sample and a list sample deliberately oversampling wealthy households to better capture the skewed wealth distribution ([Barceló, 2006](#)). The SCF is a benchmark for wealth imputation research due to its detailed financial data and the known complexities arising from its design and the nature of wealth. Item nonresponse in public-use SCF datasets is addressed by the Federal Reserve through a multiple imputation approach that generates five complete datasets with different imputed values, using sequential regression-based procedures that incorporate range constraints, logical data structures, and empirical residuals to preserve the complex multivariate relationships inherent in wealth data ([Kennickell, 1998](#)).

Specifically, we use the 2022 summarized SCF as our donor dataset.

### 3.2 Current Population Survey

The Current Population Survey (CPS), conducted jointly by the U.S. Census Bureau and the Bureau of Labor Statistics, is a monthly survey of approximately 60,000 U.S. households that serves as the primary source of labor force statistics for the United States. The CPS uses a multistage probability-based sample designed to represent the civilian non-institutional population; on average, each sampled household represents approximately 2,500 households in the population. The Annual Social and Economic Supplement (ASEC) extends the core survey with detailed annual income data, including earnings, unemployment compensation, Social Security, pension income, interest, dividends, and other income sources. However, despite its comprehensive coverage of income and employment, the CPS does not collect information on household wealth, assets, or liabilities, which are key variables needed for wealth-based policy analysis. This omission motivates the need for statistical matching to transfer wealth information from the SCF onto the CPS.

### 3.3 Comparative analysis and characteristics for imputation

Imputing between SCF and CPS exemplifies the challenges of statistical matching, as it requires combining a specialized survey with detailed wealth data but limited sample size with a large representative survey that lacks wealth measures entirely. The SCF-to-CPS imputation is directly relevant to U.S. policy microsimulation, where researchers require comprehensive microdata combining income, demographics, and wealth to analyze the distributional effects of tax and benefit reforms. By imputing wealth onto the CPS, analysts can examine wealth distributions in a sample roughly thirteen times larger than the SCF alone, enabling finer demographic disaggregations and more precise subgroup analyses. Moreover, the heavy-tailed nature of wealth distributions and the different sampling designs of these surveys create methodological challenges that make this an informative test case for comparing imputation methods.

## 4 Methodology

### 4.1 microimpute package implementation

`microimpute`<sup>1</sup> is a Python framework that enables variable imputation through multiple statistical methods, providing a consistent interface for comparing and benchmarking different imputation approaches using quantile loss calculations.

#### 4.1.1 Core capabilities

The package currently supports four primary imputation methods: Hot Deck Matching, Ordinary Least Squares Linear Regression, Quantile Regression Forests (QRF), and Quantile Regression. This approach allows researchers to systematically evaluate which technique provides the most accurate results for their specific dataset and research objectives. Additionally, the package is designed to be modular, allowing for easy extension with additional imputation methods in the future.

#### 4.1.2 Key features for microimputation

`microimpute` addresses imputation challenges between complex survey datasets through several specialized features:

1. **Survey data weights integration:** Handles survey data weights through sampling to ensure that models are trained on a donor data set representative of the true distribution.
2. **Method comparison and benchmarking:** Allows researchers to easily compare different approaches and automatically determine the method providing the most accurate results.
3. **Flexible methodological set-up:** Enables advanced usage through specified hyperparameter setting and tuning.
4. **Quantile-based evaluation:** Uses quantile loss calculations to assess imputation quality across different parts of the distribution.
5. **Autoimputation:** Provides an integrated imputation pipeline that tunes method hyperparameters to the specific datasets, compares methods, and selects the best-performing to conduct the requested imputation in a single function call.

### 4.2 Evaluation Framework

Comparing imputation methods requires metrics that assess distributional accuracy, not just point prediction. The `microimpute` package uses quantile loss as the primary evaluation metric:

$$\mathcal{L}_\tau = \rho_\tau(y - \hat{Q}_\tau(y|x)) = (y - \hat{Q}_\tau(y|x)) \cdot (\tau - \mathbf{1}_{y < \hat{Q}_\tau(y|x)}) \quad (12)$$

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<sup>1</sup>Complete documentation, implementation details, and usage examples are available at <https://policyengine.github.io/microimpute/>.

Averaging over multiple quantiles  $\tau \in \{0.1, 0.2, \dots, 0.9\}$  provides a comprehensive measure of how well the method estimates the conditional distribution across its entire range. Lower average quantile loss indicates better distributional calibration.

Additional metrics include:

- **Wasserstein distance:** Measures the distance between the marginal distributions of observed and imputed values
- **Kolmogorov-Smirnov statistic:** Tests whether observed and imputed values come from the same distribution

These metrics enable systematic comparison across methods, allowing researchers to select the approach best suited to their specific data characteristics.

## 5 Results

### 5.1 Evaluation metrics

To properly assess imputation quality for wealth data, we employ quantile loss to evaluate performance across quantiles. Quantile (or pinball) loss evaluates how well a model predicts a chosen conditional quantile by assigning an asymmetric penalty that depends on the sign of the forecast error and the target quantile (Koenker and Bassett, 1978). Formally, for residual  $e_i = y_i - \hat{y}_i$  and quantile level  $\tau \in (0, 1)$ , the loss is

$$L_\tau(e_i) = \max(\tau e_i, (\tau - 1) e_i);$$

thus, under-prediction of an upper-tail quantile (large positive  $e_i$ ) is penalized at rate  $\tau$ , whereas over-prediction is penalized at rate  $\tau - 1$ . Minimising this piece-wise linear function ensures that, in expectation, exactly  $\tau$  percent of true outcomes fall below the model’s prediction, giving the estimator its distribution-calibrating property first formalised by Koenker and Bassett (Koenker and Bassett, 1978). Because the weights differ for positive versus negative errors, quantile loss captures directional bias. This is a crucial feature when imputing highly skewed variables like wealth, where large under-estimates in the right tail must be discouraged more strongly than equal-sized over-estimates. Compared with mean-squared or mean-absolute error, which optimise the conditional mean or median and penalise errors symmetrically, quantile loss directly targets the entire conditional distribution and remains robust to outliers, making it the appropriate metric for assessing the performance of imputation models when distribution-awareness is a priority (Ghenis, 2018).

Additionally, we ensure that the resulting wealth distributions resemble that of the donor wealth distribution in the SCF. The SCF includes household weights to quantify how representative each recorded household and its data are. Thus, by sampling the SCF distribution according to its weights, we can get an accurate estimate of the true US wealth distribution, which simultaneously serves as a visual check for imputed wealth distributions.

Lastly, by evaluating the average wealth by disposable income decile that results from each method’s imputations, we can compare whether and how sensibly wealth imputations correlate with income as expected for each method’s results, with households that have the highest disposable income also having the highest wealth.

## 5.2 Experimental setup

Using Microimpute’s `autoimpute` function, we evaluate QRF against OLS, standard Quantile Regression, and hot Deck Matching in net worth imputation. To create a ground truth for evaluation, we employ cross-validation, splitting the SCF into 5 folds, or subsets. For each model at a time, we:

1. Mask the wealth values of one holdout subset at a time
2. Tune the hyperparameters of those models that have a flexible set-up to the SCF data set, namely Matching<sup>2</sup> and QRF<sup>3</sup>
  - (a) Hyperparameter tuning is done through the `optuna` package, which paralelizes hyperparameter searches to identify the combination of hyperparameters that minimizes the average quantile loss across all quantiles for the SCF dataset.
3. Impute wealth (`networth`) onto the SCF holdout set using each method with the remaining four folds as the training data
  - (a) SCF survey data weights (captured in the variable `household_weight`) are passed into `autoimpute` for sampling
  - (b) Demographic and financial variables available in both surveys behave as predictors, after being preprocess for consistent variable names and encoding: `is_female`, `age`, `race`, `own_children_in_household`, `employment_income`, `interest_dividend_income`, `pension_income`
  - (c) Each model performs imputation at 20 equally spaced quantiles
4. Compare the imputed values to the SCF training values, measuring quantile loss at each of the imputation quantiles
5. Average quantile loss across all folds and quantiles

This approach allows for direct comparison between methods on a common testing framework, providing a robust assessment of relative performance. The model with the lowest average quantile loss is chosen as the best performing and automatically selected to perform the final imputation onto the CPS data set, for which the full SCF will be used for training.

Next, the imputation from the SCF onto the CPS is replicated for the other three models that were not selected, with an identical setup that includes training on the full SCF, the same predictor variables, and imputations at the median quantile, avoiding introducing any bias toward a specific side of the distribution.

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<sup>2</sup>hyperparameters: `dist_fun`, `constrained`, `constr_alg`, `k`

<sup>3</sup>hyperparameters: `n_estimators`, `min_samples_split`, `min_samples_leaf`, `max_features`, `bootstrap`

## 5.3 Imputation results

The cross-validation results demonstrate QRF’s superior performance across the wealth distribution. QRF achieved an average quantile loss of 6.6m across all quantiles, outperforming OLS (8.3m), Hot Deck Matching (7.75m), and standard Quantile Regression (7.05m). This 20.5% improvement over OLS and 14.8% over Matching and 6.4% improvement over Quantile Regression represents a substantial gain in imputation accuracy.

The performance advantage was particularly pronounced in the middle portions of the distribution (10th-80th percentiles), where QRF maintained consistently lower quantile loss values. While Quantile Regression showed competitive performance at extreme quantiles (above the 85th percentile), QRF’s overall consistency across the entire distribution made it the optimal choice for wealth imputation.

Microimpute’s automated hyperparameter tuning contributed significantly to these results. The optimal QRF configuration identified through cross-validation included 200 trees and a minimum of 5 samples per leaf node for a split. These parameters balanced model complexity with generalization capability, preventing overfitting while capturing the nuanced relationships between demographic predictors and wealth outcomes.

### 5.3.1 Quantile loss

Comparing all four models to each other, we can evaluate performance through quantile loss, as well as comparing the overlap of the imputed wealth distribution with the original SCF data.

We observe how QRF proves to be the best-performing model out of the four. It significantly outperforms Matching and OLS on average, while only being outperformed by QuantReg past the 80th quantile. Matching’s performance presents the perfect opportunity to see the quantile loss’s directional bias at work. Because Matching does not incorporate quantile information into its imputation process in any way, it will match donor and receiver units, and thus impute values identically regardless of the quantile at work. The laddering behavior observed above results from the fact that Matching’s wealth imputations consistently underpredict relative to the true wealth values, and this property is increasingly favored for quantiles below the median, and increasingly penalized as quantiles approach the right end of the distribution.

### 5.3.2 Imputed wealth distribution comparisons

By visually comparing the wealth distributions resulting from imputing with each method, and comparing them to the weighted donor distribution, we gain a more comprehensive understanding of imputation performance, moving past the test quantile loss average measured on the SCF. The distribution of wealth values imputed by QRF closely resembles the original SCF distribution, suggesting that QRF is not only the best method in terms of quantile loss but also strong when achieving distributional estimates close to the true wealth distribution in the United States. Because Matching directly takes values from the donor distribution and imputes them onto the receiver, it is unsurprising that its distribution closely resembles the donor distribution. However, Matching’s performance in quantile loss confirms that even

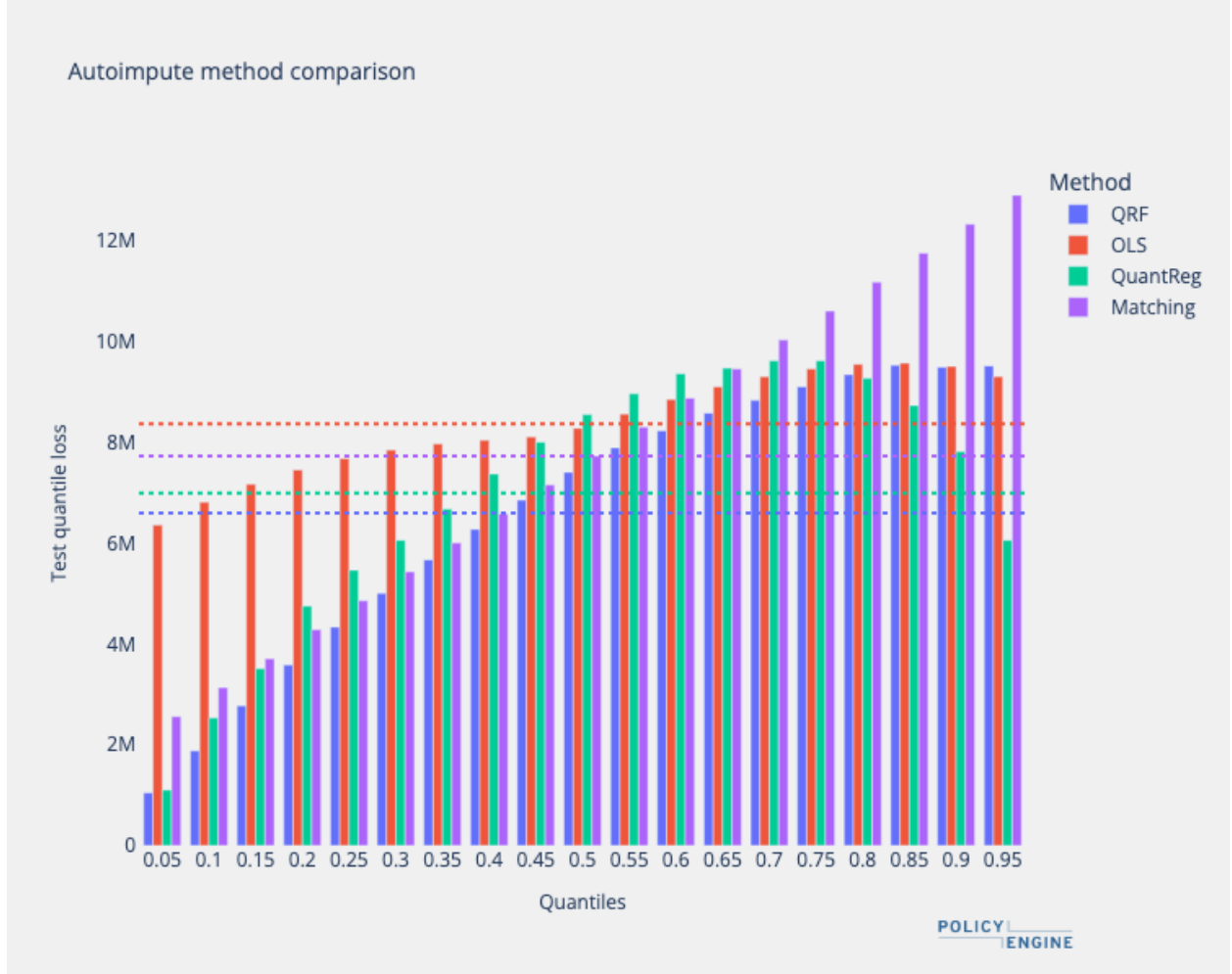


Figure 1: Cross-validation method performance at 20 equally spaced quantiles. Test quantile loss at each quantile is averaged across 5 cross-validation folds, and the dashed lines represent average quantile loss across all quantiles.

if it takes the exact values from the donor, it may struggle to uncover the non-linear relationship between the different predictors and wealth values, resulting in a seemingly correct data distribution but innacurate imputation given certain household characteristics. Meanwhile, OLS and QuantReg fail to capture the variability across the distribution and impute most values at the lower or higher ends of the distribution.

### 5.3.3 Distribution of wealth by disposable income decile

These results support the observations above, with QRF presenting the most consistent and plausible relationship to disposable income, with a gradually increasing average as the deciles increase. This plot also demonstrates the caveats of the other models, for example showing the lower variability in OLS's predictions, and Matching's consistent underprediction across deciles.

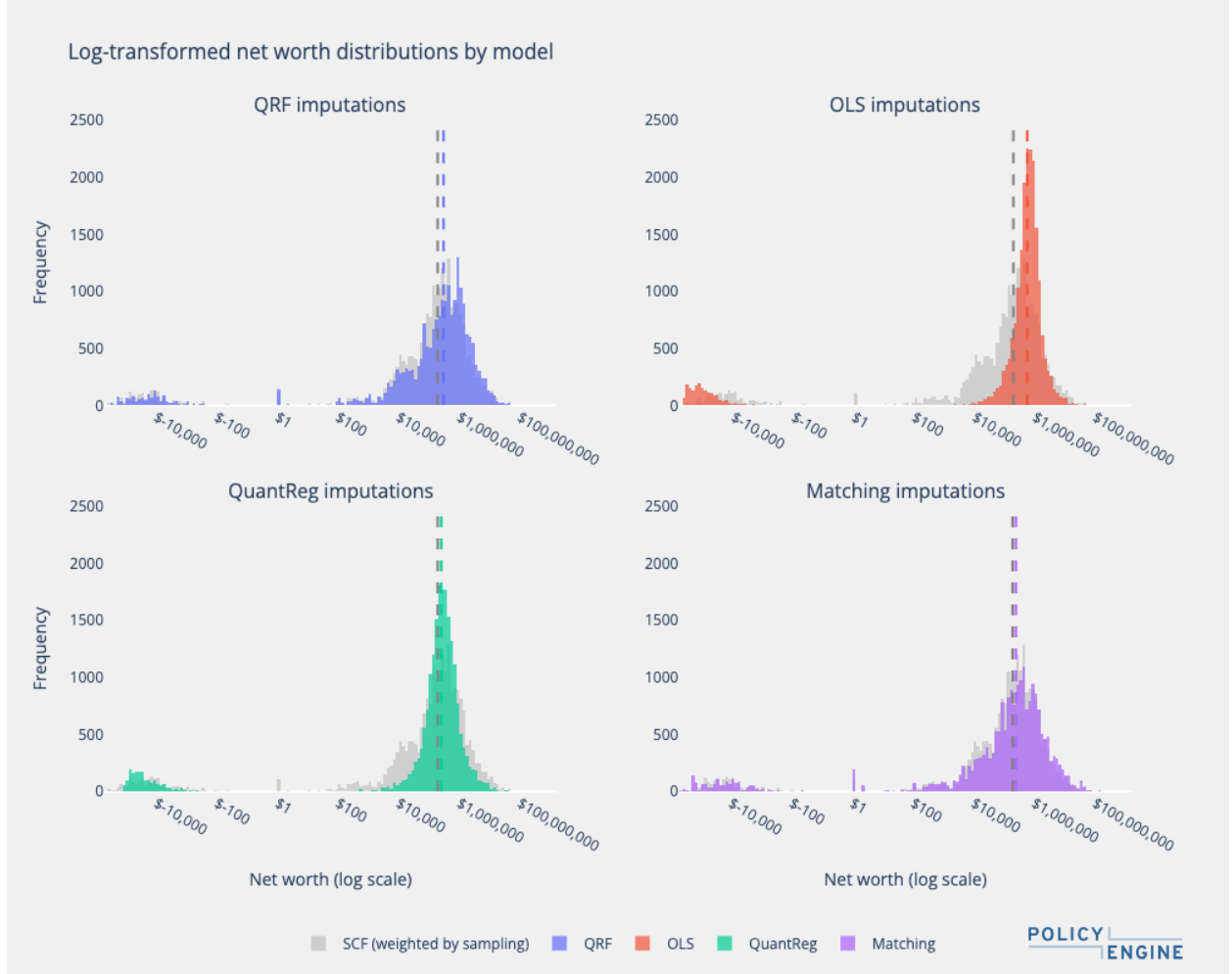


Figure 2: Weight-adjusted imputed wealth distributions model comparisons, relative to the donor SCF wealth distribution. Dashed lines represent median values.

## 6 Discussion

This paper has demonstrated both theoretically and empirically that Quantile Regression Forests provide substantial advantages for microimputation, particularly through the example of wealth imputation from the SCF to the CPS. By preserving the full conditional distribution of wealth, QRF maintains the critical statistical properties of wealth data that traditional methods fail to capture.

### 6.1 Strengths powered by microimpute

The `microimpute` package’s design philosophy and implementation choices provide several key advantages that contributed to the success of our wealth imputation analysis:

1. **Unified interface for method comparison:** `microimpute`’s consistent API across all imputation methods enabled systematic benchmarking without implementation-



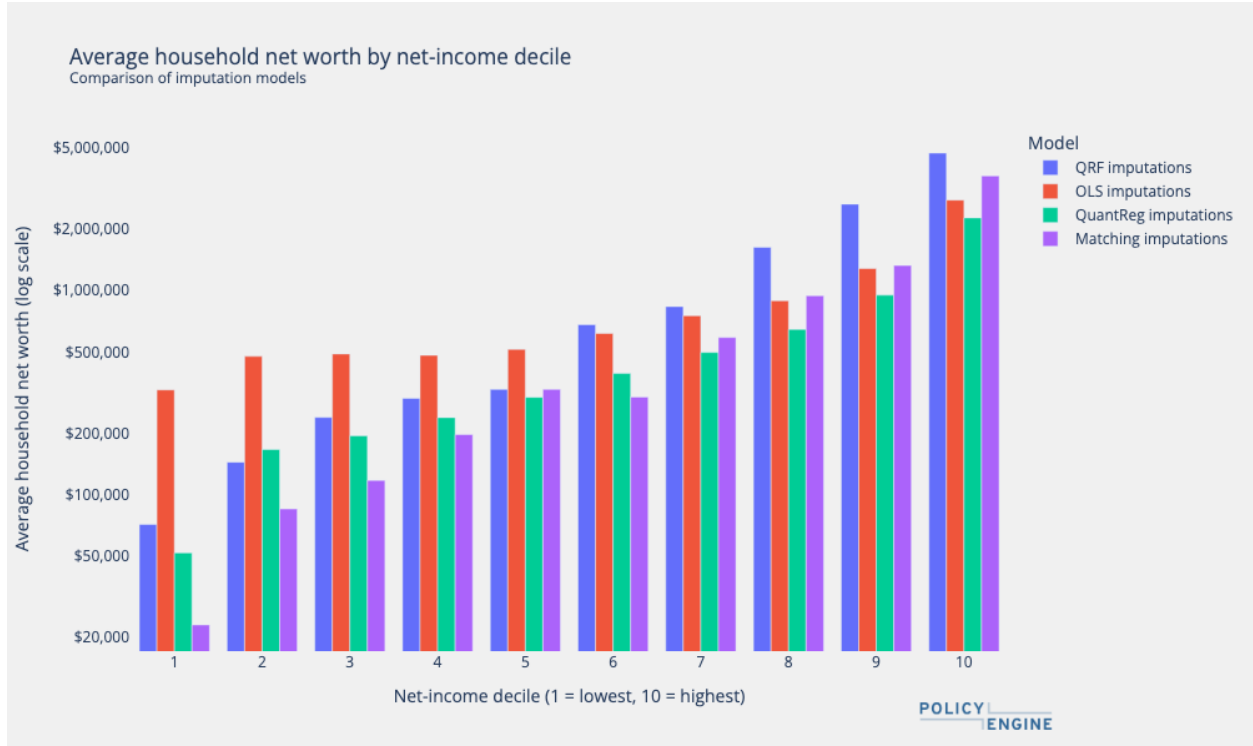


Figure 3: Average household net worth by disposable income decile model comparison. Households are divided into 10 equally sized groups based on their disposable income.

specific biases. This standardization ensures that performance differences reflect genuine methodological advantages rather than implementation artifacts (PolicyEngine, 2025).

2. **Automated method selection:** The package’s `autoimpute` function streamlines the imputation workflow by automatically comparing methods and selecting the best performer based on quantile loss metrics. This feature proved particularly valuable given the wealth data’s complexity, as it removed subjective method selection and ensured optimal performance.
3. **Survey weight integration:** `microimpute`’s native support for survey weights through stratified sampling ensures that imputation models properly represent population distributions. This capability is crucial when transferring information between surveys with different sampling designs, such as the SCF’s oversampling of wealthy households.
4. **Quantile-aware evaluation:** By implementing quantile loss as the primary evaluation metric, `microimpute` directly addresses the challenges of skewed distributions. This metric’s asymmetric penalty structure naturally prioritizes accurate imputation at distribution tails, where traditional metrics like Root Mean Squared Error (RMSE) often fail.
5. **Computational efficiency:** The package’s optimized implementation enables processing of large microdata files while maintaining reasonable computation times. Cross-

validation on the full SCF dataset, including QRF hyperparameter tuning, was completed in under 30 minutes on standard hardware, making iterative experimentation feasible.

6. **Open-source transparency:** As an open-source tool, `microimpute` allows full inspection and modification of imputation algorithms, promoting reproducibility and enabling custom extensions for specific research needs ([PolicyEngine, 2025](#)).

## 6.2 Limitations and future improvements

Despite the demonstrated advantages, several limitations warrant consideration for future development:

### 6.2.1 Current package limitations

While `microimpute` currently supports four imputation methods, expanding to include modern machine learning approaches such as neural networks, gradient boosting machines, or deep learning architectures could further improve performance, particularly for complex multivariate relationships ([Alaa et al., 2024](#)). The package would benefit from implementing ensemble methods that combine multiple imputation approaches, potentially leveraging the strengths of different methods across different parts of the distribution. Moreover, model selection and assessment could be enhanced with evaluation metrics additional to quantile loss, ensuring a thorough understanding of each model’s behavior at every step.

### 6.2.2 QRF-specific challenges

The terminal node sparsity issue identified in Section 2.2.4 remains a fundamental limitation of tree-based methods. When few training samples reach certain terminal nodes, multiple observations in the recipient dataset may receive identical imputed values, potentially underestimating variability in extreme wealth categories. Future work could explore adaptive tree construction methods that ensure minimum node occupancy or hybrid approaches that combine QRF with parametric methods at distribution extremes.

These enhancements would position `microimpute` further as a comprehensive solution for survey data imputation challenges while maintaining its current strengths in ease of use and methodological rigor.

## 7 Conclusion

This paper has reviewed methodological advances in microdata imputation, focusing on techniques suitable for complex survey data like wealth. We highlighted the limitations of traditional methods and the advantages of novel approaches, particularly Quantile Regression Forests, for handling skewed distributions and non-linearities.

The key contributions of this work are the synthesis of current knowledge on imputation for challenging microdata, a detailed examination of QRF’s suitability, and an introduction to the `microimpute` package. Our review suggests that QRF represents a step forward

in preserving the statistical integrity of imputed microdata, increasing the robustness of economic analysis. The implementation of QRF in the `microimpute` package provides a practical tool for researchers seeking to combine detailed microdata across datasets.

Future research should continue to refine QRF for imputation, particularly in response to challenges like limited data at extreme quantiles. Comparative studies against other emerging techniques, like deep learning models (Alaa et al., 2024), are also vital. Continued innovation in imputation methodology supports the reliability of evidence-based research and policymaking.

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